

Highly efficient and tunable polarization rotator based on lithium niobate on an insulator

XIANGYU HE,¹ YIN XU,^{1,2}  BO ZHANG,^{1,2} YUE DONG,^{1,2}  AND YI NI^{1,2,*}

¹Department of Electronic Engineering, School of IoT Engineering, Jiangnan University, Wuxi 214122, China

²Institute of Advanced Technology, Jiangnan University, Wuxi 214122, China

*8073110160@jiangnan.edu.cn

Received 8 December 2022; revised 12 February 2023; accepted 13 February 2023; posted 15 February 2023; published 21 March 2023

The lithium niobate on an insulator (LNOI) platform has greatly advanced the development of integrated photonics recently, where efficient polarization management components are indispensable. In this work, we propose a highly efficient and tunable polarization rotator based on the LNOI platform and the low-loss optical phase change material antimony triselenide (Sb_2Se_3). The key polarization rotation region is formed by a LNOI waveguide with a cross section of the double trapezoidal shape and a Sb_2Se_3 layer deposited atop the LNOI waveguide in an asymmetrical way, where an isolating layer of silicon dioxide is sandwiched between them to reduce the material absorption loss. Based on such a structure, we have achieved the efficient polarization rotation in a length of only 17.7 μm , where the polarization conversion efficiency and insertion loss are 99.6% (99.2%) and 0.38 dB (0.4 dB) for the trans-electric (TE)-to-trans-magnetic (TM) rotation. If we further change the phase state of the Sb_2Se_3 layer, other polarization rotation angles besides 90° can also be obtained for the same device, revealing a tunable function. We believe that the proposed device and design scheme could offer an efficient method for realizing the polarization management on the LNOI platform. © 2023 Optica Publishing Group

<https://doi.org/10.1364/AO.483171>

1. INTRODUCTION

Lithium niobate (LN) crystal is a synthetic material with excellent piezoelectric, electro-optic, acousto-optic, and nonlinear features with quite stable physical and chemical properties, which has been widely used in modulators and nonlinear devices of optical communications [1–3]. However, such a bulk crystal material and related devices are difficult to integrate with other components on the chip to form the photonic integrated circuits (PICs). Fortunately, thin-film LN has been realized via using a smart cut fabrication process, and the corresponding lithium niobate on insulator (LNOI) wafer can be easily available from commercial companies at present [4,5]. The most significant advantage of the LNOI wafer is its relatively high refractive-index-contrast ($\Delta n > 0.7$), which can support small waveguides and compact devices integrated on the same chip. Meanwhile, the excellent properties of LN can be well kept for the LNOI wafer nearly without deterioration. Therefore, the LNOI has become a vital important platform for the PICs [6–8]. Nonetheless, such a relatively high refractive-index-contrast of the LNOI wafer and the birefringence feature of the LN material will introduce obvious polarization dependence, which requires efficient polarization manipulation components [9,10]. A polarization rotator (PR), a key polarization management component, is employed to achieve the polarization conversion between different polarization states [11,12]. High

conversion efficiency (CE) and low insertion loss (IL), as well as compact size, are our desirable for the LNOI-based PR.

The PR has been extensively studied in the silicon on insulator (SOI) platform and the device working principle can be classified into two types: mode evolution and mode interference [13–17]. The mode evolution-based PR normally requires a rather long length (e.g., $>100 \mu\text{m}$) to gradually achieve the mode evolution from the input to output polarization mode [13,14], while the obtained working bandwidth and fabrication tolerance are relatively large due to the evolution process. As a contrast, the mode interference-based PR can be very short ($\sim 10 \mu\text{m}$), but at the cost of a narrow working bandwidth and tight fabrication tolerance [15–17], which might pose challenges for the device fabrication and application. For the LNOI-based PR, its working principle is similar to the above-mentioned PR based on the SOI platform. Recently, several works have been done to realize the polarization rotation based on the LNOI platform. For example, Chen *et al.* designed and fabricated a LNOI-based PR using adiabatically tapered waveguides, where the length of the waveguide taper is 4000 μm with a broad bandwidth of 500 nm [18]. Such a long conversion length might be not friendly for the on-chip compact integration. To shorten the conversion length, Zhao *et al.* proposed a rotation scheme based on the asymmetric hybrid plasmonic waveguide [19]. Using the mode interference between two hybrid modes within the conversion region, the polarization

rotation can be achieved in a length of only 13.7 μm which is greatly reduced. However, the device loss is a big problem because it is larger than 1.4 dB due to the introduced metal absorption loss and requires efficient ways to reduce such loss. In addition, other polarization manipulation devices have been reported for the LNOI platform [20–22]. By analyzing these reported PRs, we can find that their device functions are fixed when their structures are determined, which cannot be further changed. Additionally, the rotation angle of the PR is set to 90° , which cannot achieve other polarization rotation angles for the same device. Therefore, a new LNOI-based PR with a tunable function, high conversion performance, and compact size still needs to be developed.

With the optical phase change material (OPCM), the amorphous and crystalline states have clearly different optical properties. These two typical phase states can be transformed via using thermal, electrical, or optical stimulus, and every phase state can be sustained for a long time without any power supply [23–25]. Thus, the OPCM has been widely used in the optical switch [26], optical modulation [27], optical computing [28], optical neural network [29], and nonvolatile memory [30]. Compared with the commonly used OPCM, e.g., VO_2 , $\text{Ge}_2\text{Sb}_2\text{Te}_5$, and $\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}_1$, antimony triselenide (Sb_2Se_3) is a new OPCM which has quite low optical loss (imaginary part $< 10^{-5}$) in the optical communication bands even, at the crystalline state [31,32]. By sputtering such material on the LNOI waveguide, the optical mode and light transmission behavior could be tuned in a nonvolatile way.

In this paper, we propose a highly efficient and tunable PR based on the LNOI platform. The key polarization rotation region is formed by a LNOI waveguide with a cross section of the double trapezoidal shape and an extra layer of Sb_2Se_3 coated atop the LNOI waveguide on one side. Based on such an asymmetrical structure, the polarization rotation will take place along the light propagation direction. By setting the working state (crystalline state) of the Sb_2Se_3 layer and optimizing the structural parameters, we have achieved the polarization rotation from the input TE (TM) to output TM (TE) mode with CE = 99.6% (99.2%) and IL = 0.38 dB (0.4 dB) in a length of 17.7 μm . If we further change the working state of the Sb_2Se_3 layer, other polarization rotation angles can also be obtained for the same device, which could support new polarization applications. Therefore, we propose the proposed device scheme can be applied in the efficient polarization management based on the LNOI platform.

2. DEVICE STRUCTURE AND PRINCIPLE

Figure 1 shows the schematic of our proposed highly efficient and tunable PR on a 600 nm thick X-cut LNOI wafer. The input and output waveguides are based on the LNOI waveguide with a double trapezoidal shape in the cross section, where the top widths and thickness of the double trapezoidal shape are w_1 , w_2 and h_1 ($= 300$ nm), h_2 ($= 300$ nm), respectively. Within the polarization rotation region, we add a Sb_2Se_3 layer with a thickness of h_p coated at the top surface of the LNOI waveguide on one side, where a layer of silicon dioxide is sandwiched between them in order to reduce the absorption loss of the Sb_2Se_3 layer working at the crystalline state. The vertical gap

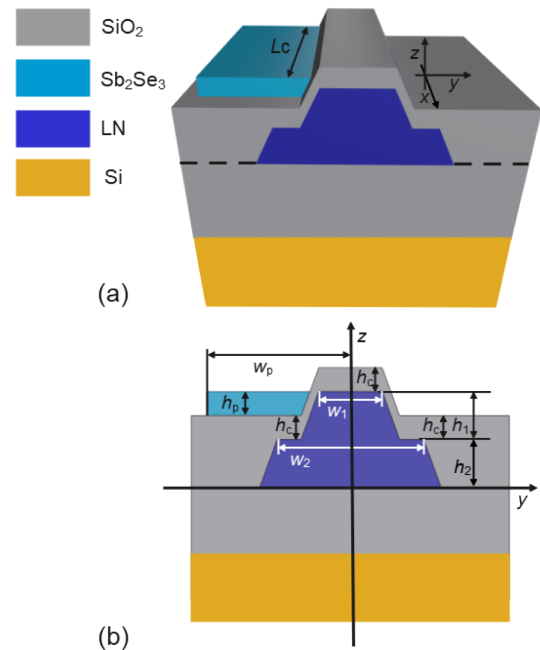


Fig. 1. Schematic of the proposed tunable PR on the LNOI platform. (a) 3D view and (b) cross-sectional view of the key polarization rotation region. The structural and material parameters are also labeled.

between the Sb_2Se_3 layer and the bottom trapezoidal waveguide is h_c , and the gap between the outside of the Sb_2Se_3 layer and the waveguide center is w_p , respectively. The polarization rotation length is set as L_C , which is the same as the length of the Sb_2Se_3 layer in the propagation direction. In addition, the sidewall angle of the LNOI waveguide is designed to be 70° according to the current etching technique for the LNOI wafer [4,5]. In this paper, we choose the x-cut LNOI wafer. The ordinary index (n_o) is along the x and z axis, and the extraordinary index (n_e) is along the y axis; n_o and n_e are 2.211 and 2.138, respectively, at the wavelength of 1550 nm. The refractive indices of SiO_2 is 1.44 at the wavelength of 1550 nm. The refractive indices of amorphous and crystalline Sb_2Se_3 are assumed as 3.28 and 4.05 at the wavelength of 1550 nm, respectively.

Based on such a device structure, the structural symmetry is clearly broken by the added Sb_2Se_3 layer. When the input mode enters into the rotation region, two hybrid modes will be excited and beaten with each other along the propagation direction, where these hybrid modes are the eigenmodes of the hybrid waveguide in the rotation region. The rotation length L_C is designed to make the phase difference between the two hybrid modes equal to π . $L_C = \pi / (\beta_1 - \beta_2)$, where β_1 and β_2 are the propagation constants of the hybrid modes [15]. After the rotation region, the two hybrid modes will be combined, forming an output polarization mode, where its polarization direction is vertical to the polarization direction of the input mode due to the accumulated phase difference of π between the hybrid modes. To evaluate the polarization rotation performance, we employ the polarization conversion efficiency (PCE), which is defined as the output power of the desired polarization mode over the sum of both polarization modes at the device output port. For example, for the TE-to-TM rotation, the

$PCE = P_{TM}/(P_{TM} + P_{TE})$, where P_{TE} and P_{TM} are the output power of the TE and TM modes at the device output port. Meanwhile, the PCE can also be expressed by the optical axis rotation angle θ of the hybrid modes [12,15]:

$$PCE = \sin^2(2\theta) \sin^2\left(\frac{\pi L}{2L_c}\right) \times 100\%, \quad (1)$$

where L is the device length.

Thus, in order to get a high PCE (e.g., $\sim 100\%$), the device length L should be equal to the rotation length L_c , and the optical axis rotation angle θ should be equal to 45° . Additionally, the optical axis rotation angle θ is defined as [12,15]

$$\tan(\theta) = \frac{\int \int_{\Omega} \varepsilon(y, z) \cdot H_y^2(y, z) dy dz}{\int \int_{\Omega} \varepsilon(y, z) \cdot H_z^2(y, z) dy dz}, \quad (2)$$

where $\varepsilon(y, z)$ is the permittivity distribution, $H_y(y, z)$ and $H_z(y, z)$ are the magnetic components of the eigenmode in the horizontal and vertical directions, and the integration region is the cross section of the hybrid waveguide. Therefore, under the following analyses, we will use Eq. (2) to determine the cross-sectional parameters of the hybrid waveguide by considering the requirement of $\theta = 45^\circ$ and then employ Eq. (1) to find the optimum PCE.

3. RESULTS AND DISCUSSION

To study the modal features of the proposed PR, a full vectorial mode solver based on the finite difference time domain method is employed [33,34]. Figure 2 shows the calculated optical axis rotation angle θ as a function of the gap width w_p and the Sb_2Se_3 layer thickness h_p , the results of which are based on Eq. (2). Note that the gap width w_p has more influence on the optical axis rotation angle θ , and θ reduces from 90° to nearly 0° when w_p increases from 600 to 900 nm. If we want to satisfy the requirement of $\theta = 45^\circ$, w_p and h_p should be designed as 800 and 150 nm, respectively. Within the calculation process, the waveguide widths are set as $w_1 = 400$ nm, $w_2 = w_1 + 2h_1/\tan 70^\circ + 300$ nm, and the vertical gap width is $h_c = 150$ nm. Moreover, we also analyze the waveguide width

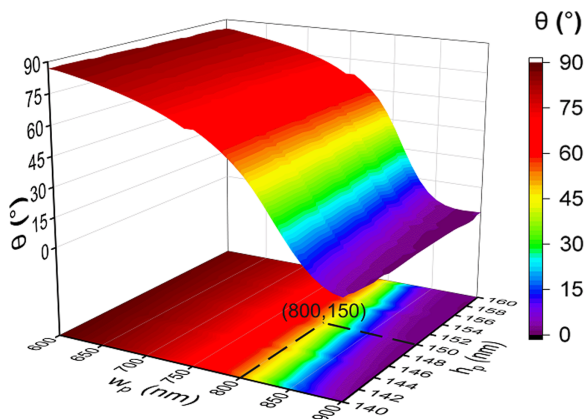


Fig. 2. Optical axis rotation angle θ as functions of the gap w_p between the outside of the Sb_2Se_3 layer and the waveguide center, and the Sb_2Se_3 layer thickness h_p . The chosen optimum parameters of w_p and h_p are also marked by satisfying the requirement of $\theta = 45^\circ$.

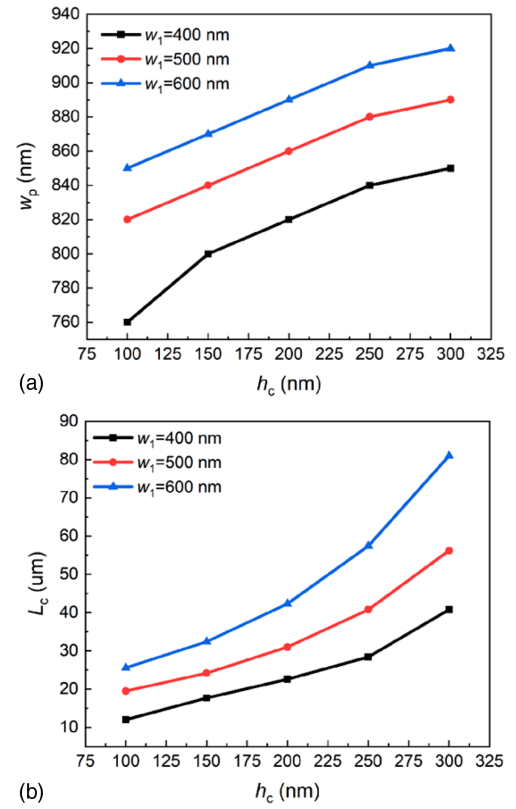


Fig. 3. (a) Relation between the gap width w_p and the vertical gap width h_c for different waveguide widths w_1 under the requirement of $\theta = 45^\circ$. (b) Corresponding polarization rotation lengths L_c as functions of the vertical gap width h_c and the waveguide width w_1 .

w_1 , vertical gap width h_c , and gap width w_p under the requirement of $\theta = 45^\circ$, as shown in Fig. 3(a). We can find that w_p will increase as w_1 and h_c increase and the reason is that the required Sb_2Se_3 layer width should be increased when the waveguide size is enlarged. Meanwhile, we also calculate the corresponding polarization rotation length L_c shown in Fig. 3(b). From Fig. 3(b), we can observe that L_c gradually increases as the waveguide size is enlarged. By analyzing these results, the waveguide width w_1 is set to 400 nm, and the corresponding h_c and L_c are 150 nm and $\sim 18 \mu\text{m}$. Therefore, according to the results obtained from Figs. 2 and 3, the cross-sectional parameters of the proposed PR are determined according to the requirement of $\theta = 45^\circ$.

Next, we perform the device transmission analyses using the mature three-dimensional finite difference time domain method [35]. Figure 4 illustrates the calculated PCE and IL of the proposed PR as a function of the device length L for the TE-to-TM and TM-to-TE rotation. By overall consideration, the optimum device length locates within the range from 17.4 to $18 \mu\text{m}$, the value of which is similar to the result ($\sim 18 \mu\text{m}$) based on the modal analysis. Here we set the device length to $17.7 \mu\text{m}$, and the PCE and IL are 99.6% (99.2%) and 0.38 dB (0.4 dB), respectively, for the TE-to-TM (TM-to-TE) rotation. If the device length varies $\pm 0.3 \mu\text{m}$ from the designed value ($17.7 \mu\text{m}$), the device performance will reveal no obvious deterioration, providing a relatively large tolerance for the practical device fabrication. Moreover, we also conduct the spectrum

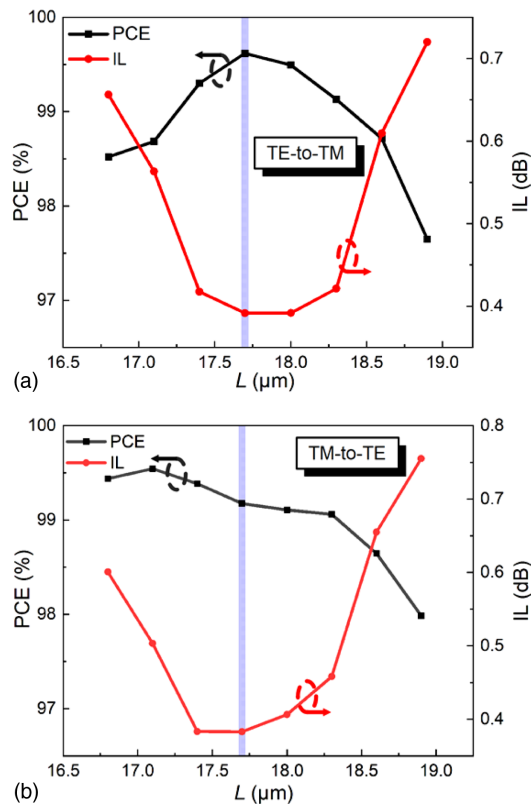


Fig. 4. Calculated PCE and IL of the proposed PR as a function of the device length L for the (a) TE-to-TM and (b) TM-to-TE rotation. The optimum device length is set to $17.7 \mu\text{m}$.

analyses of the device performance for the proposed PR under the conditions of TE-to-TM and TM-to-TE rotation when material dispersions are considered [36], as shown in Fig. 5. Within the calculated wavelength range from 1.5 to $1.66 \mu\text{m}$, both polarization rotations reveal strong wavelength dependence. If we set the criterion of the $\text{PCE} > 96\%$, the allowable wavelength ranges can be varied from 1508 to 1615 nm for TE-to-TM rotation and from 1507 to 1642 nm for TM-to-TE rotation. Meanwhile, the obtained ILs are lower than 1 dB within these wavelength ranges. Over 100 nm for the working bandwidth of the proposed PR could support the on-chip broadband applications well.

Figure 6 shows the magnetic field evolution along the propagation direction through the proposed PR, where $w_1 = 400 \text{ nm}$, $w_2 = 920 \text{ nm}$, $h_c = 150 \text{ nm}$, $w_p = 800 \text{ nm}$, $h_p = 150 \text{ nm}$, $L = 17.7 \mu\text{m}$, and $\lambda = 1.55 \mu\text{m}$. As the input TE mode enters into the polarization rotation region, it's the dominant component H_x gradually fades, while its minor component H_y is excited and gradually becomes dominant. Thus, the dominant component is changed from H_x to H_y through the proposed PR, corresponding to a 90° rotation of the input TE polarization mode. Finally, we can get the TM mode at the device output. The conversion behavior of the input TM mode is similar to that of the input TE mode, and the dominant component is changed from H_y to H_x through the proposed PR, corresponding to a 90° rotation of the input TM polarization mode. Therefore, both polarization modes can be efficiently rotated through the proposed PR in a length of $17.7 \mu\text{m}$.

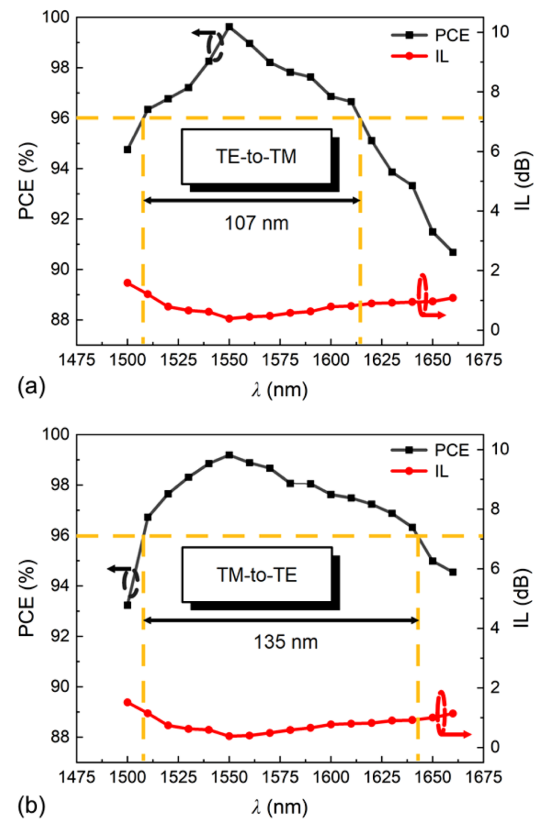


Fig. 5. Wavelength spectra of the device performance (PCE and IL) for the proposed PR under the conditions of (a) TE-to-TM and (b) TM-to-TE rotation. The working bandwidths are 107 and 135 nm , respectively, for the TE-to-TM and TM-to-TE rotation under the requirement of the $\text{PCE} > 96\%$.

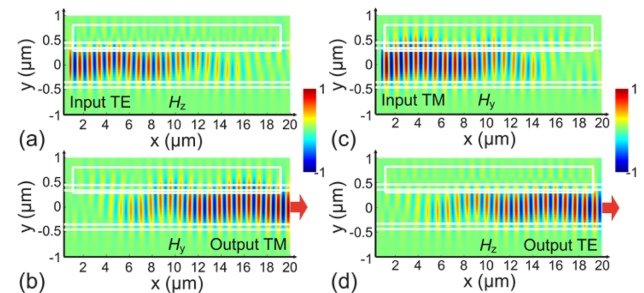


Fig. 6. Magnetic field evolution along the propagation direction through the proposed PR. (a) H_x component and (b) H_y component for the input TE mode. (c) H_y component and (d) H_x component for the input TM mode. The polarization rotation length is $17.7 \mu\text{m}$.

Considering the device fabrication, the main fabrication process can be divided into three parts: fabricating the LNOI waveguide with a double trapezoidal shape, depositing an isolating layer of silicon dioxide, and sputtering a Sb_2Se_3 layer atop the device. We can start from a 600 nm thick X-cut LNOI wafer. First, the double-trapezoidal-shaped LNOI waveguide with top widths of $w_1 = 400 \text{ nm}$, $w_2 = 920 \text{ nm}$, and thicknesses of $h_1 = 300 \text{ nm}$, $h_2 = 300 \text{ nm}$ is fabricated using E-beam lithography and reactive ion etching processes [4,5]. Secondly, an isolating layer of silicon dioxide is deposited to cover the LNOI waveguide using plasma enhanced chemical vapor deposition.

Thirdly, a Sb_2Se_3 layer is added on the top surface of the silicon dioxide using magnetron sputtering [31,32], where the position of the Sb_2Se_3 layer locates on one side of the LNOI waveguide, and the layer thickness is $h_p = 150$ nm, together with a gap $w_p = 800$ nm. Within the fabrication process, the Sb_2Se_3 layer is fabricated using the magnetron sputtering, which is a physical vapor deposition method relying on the target materials. The Sb_2Se_3 layer thickness can be controlled through the sputtering time. After the material sputtering, E-beam lithography, the reactive ion etching process should be used again to formed the required shape for the coated Sb_2Se_3 layer. Using these fabrication steps, the designed PR could be achieved. Moreover, to study the deterioration of the device performance caused by fabrication imperfections in practice, we mainly analyze the effect of the variations of the sizes of the gap w_p between the outside of the Sb_2Se_3 layer and the waveguide center, and the Sb_2Se_3 layer thickness h_p effect on the device performance, as shown in Fig. 7. The reason is that the added Sb_2Se_3 layer is the key point to achieving the polarization rotation for the present device. Here we choose TE-to-TM rotation as an example in Fig. 7. From Fig. 7, the device performance will reveal obvious deterioration when w_p and h_p shift from their designed values of $w_p = 800$ nm and $h_p = 150$ nm. If we set the same criterion of the $\text{PCE} > 96\%$, w_p and h_p should be changed within the ranges of [788.5, 811.5 nm] and [142, 154.5 nm], respectively. If the criterion of the PCE reduces from $> 96\%$ to $> 92\%$, w_p and h_p can be changed within the

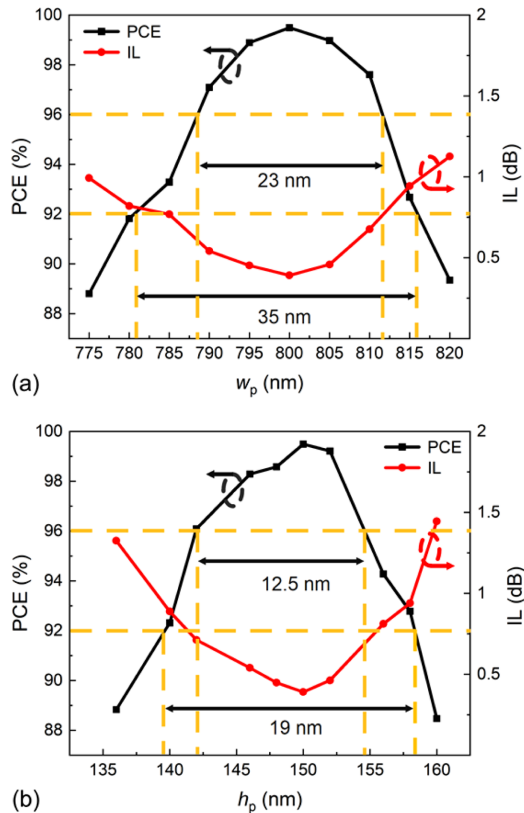


Fig. 7. Fabrication tolerance analyses of the added Sb_2Se_3 layer for the proposed PR. Calculated PCE and IL as functions of the (a) gap w_p and (b) Sb_2Se_3 layer thickness h_p . The allowable ranges for w_p and h_p are marked by keeping the $\text{PCE} > 96\%$ and the $\text{PCE} > 92\%$, respectively.

ranges of [781, 816 nm] and [139.5, 158.5 nm], respectively. The obtained ILs are always lower than 1 dB within these ranges. Note that these allowable ranges for the key parameters will offer effective guidance for the practical device fabrication.

We further study the working bandwidths of the device performance as these structural parameters (w_p and h_p) shift from their designed values, as shown in Figs. 8 and 9. From the results, we can find that the allowable working bandwidths are around 100 nm for TE-to-TM rotation (130 nm for TM-to-TE rotation) and, by keeping the $\text{PCE} > 96\%$, even the gap width w_p shifts from -10 to 10 nm relative to the designed value $w_p = 800$ nm. By contrast, the variation of the layer thickness h_p has a relatively high influence on the device working bandwidth. Within the calculation range from $h_p = 144$ nm to $h_p = 156$ nm, the largest bandwidth locates at $h_p = 150$ nm for both TE-to-TM and TM-to-TE rotation. The corresponding working bandwidth values have also been marked in Figs. 8 and 9. These obtained working bandwidths could offer useful guidance for the practical device applications.

The above-analyzed PR can efficiently achieve the 90° rotation for both the TE and TM modes on the LNOI platform, the function of which is similar to other reported PRs. Next, we will demonstrate the new tunable function of the proposed PR; such a function most likely cannot be achieved for the same device using previously reported schemes [13–19]. However, in this work, we employ the OPCM (Sb_2Se_3) to achieve the polarization rotation on the LNOI platform. When the phase

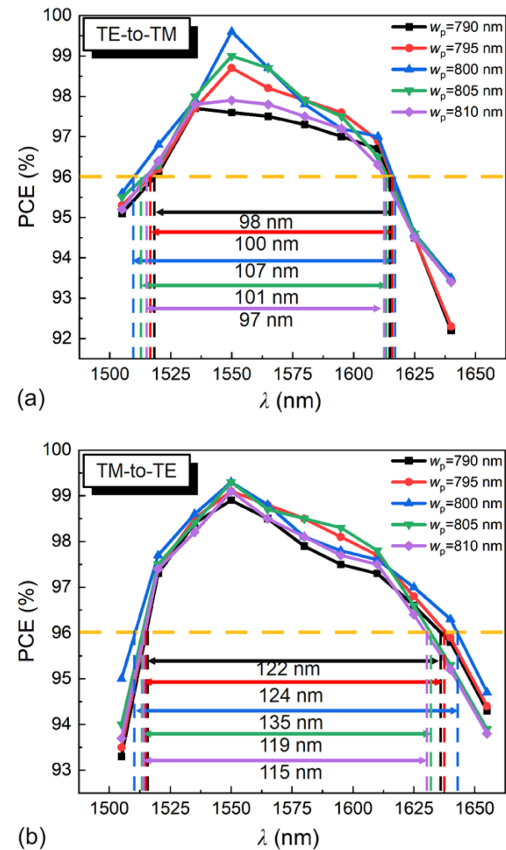


Fig. 8. Wavelength spectra of the PCE for the proposed PR as the gap width w_p shifting from its designed value ($w_p = 800$ nm) for (a) TE-to-TM and (b) TM-to-TE rotation. The allowable working bandwidths have been marked by keeping the $\text{PCE} > 96\%$.

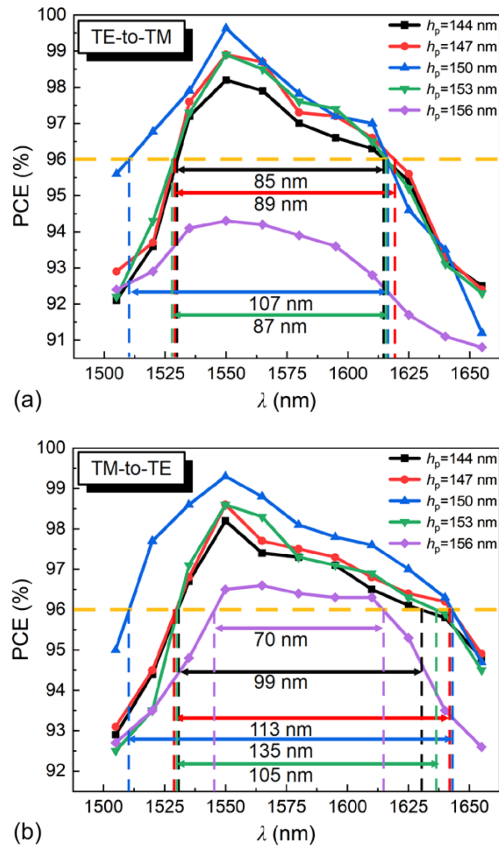


Fig. 9. Wavelength spectra of the PCE for the proposed PR as the layer thickness h_p shifting from its designed value ($h_p = 150$ nm) for (a) TE-to-TM and (b) TM-to-TE rotation. The allowable working bandwidths have been marked by keeping the PCE $> 96\%$.

state of the OPCM changes from crystalline to amorphous state, the output polarization mode will also change, revealing a new tunable function. The mechanism behind such a tunable device function is the OPCM (Sb_2Se_3) that is used. Sb_2Se_3 material can be triggered either electrically (through thermal conduction heating or electrical threshold switching) or optically.

The dielectric constant ε_{eff} of Sb_2Se_3 material is closely related to the material crystallization ratio η , which is defined as [37]

$$\frac{\varepsilon_{\text{eff}}(\lambda) - 1}{\varepsilon_{\text{eff}}(\lambda) + 2} = \eta \times \frac{\varepsilon_c(\lambda) - 1}{\varepsilon_c(\lambda) + 2} + (1 - \eta) \times \frac{\varepsilon_a(\lambda) - 1}{\varepsilon_a(\lambda) + 2}, \quad (3)$$

where ε_c is the dielectric constant of Sb_2Se_3 material in a completely crystalline state, and ε_a is the dielectric constant of Sb_2Se_3 material in a completely amorphous state.

The refractive indices of Sb_2Se_3 material in these two states are 3.64 and 3.49. Using Eq. (3), we can get the dielectric constant of Sb_2Se_3 material in the intermediate state. By adding these intermediate states of Sb_2Se_3 material into the proposed PR, the polarization rotation angle of the output mode varies, rather than staying at a fixed 90° , as reported in some previous works [13–19]. Figure 10 plots the dielectric constant ε_{eff} of Sb_2Se_3 material and the polarization rotation angle θ_m of the output mode for the proposed PR as a function of the material crystallization ratio η , where the input TE mode is employed as an example. Note that ε_{eff} is linearly increased as η increases

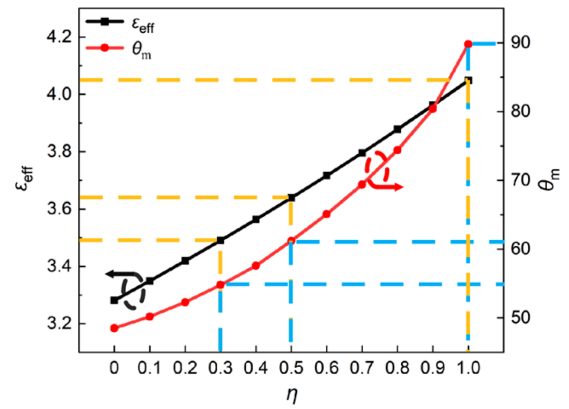


Fig. 10. Dielectric constant ε_{eff} of Sb_2Se_3 material and the polarization rotation angle θ_m of the output mode for the proposed PR as a function of the material crystallization ratio η . The input TE mode is employed during the device analysis.

from 0 to 1, where the corresponding θ_m is gradually increased from 48° to 90° . The crystallization ratio of Sb_2Se_3 material can be controlled by the width, power, and irradiation time of the laser pulse. If we can precisely control the crystallization ratio of Sb_2Se_3 material through experimental methods, the arbitrary output polarization rotation angle within the range from 48° to 90° could be obtained for the same device in theory. For example, the above-analyzed PR corresponds to $\eta = 1$, and the output polarization rotation angle is 90° relative to the input TE mode, corresponding to the output TM mode.

Here we choose $\eta = 0.5$ and $\eta = 0.3$ as examples. Figure 11 shows the magnetic field evolution of the input TE mode through the proposed PR, as the crystallization ratio η of Sb_2Se_3 material is set to 0.5 and 0.3, where the corresponding output modes are also illustrated. From Fig. 11, obvious polarization rotation behavior can be observed, and the output mode is a hybrid mode, rather than a pure TE or TM mode, where the polarization rotation angles are 61° and 55° relative to the input TE mode for $\eta = 0.5$ and $\eta = 0.3$, respectively. Other polarization rotation angles can also be achieved only by setting the required intermediate state of Sb_2Se_3 material according

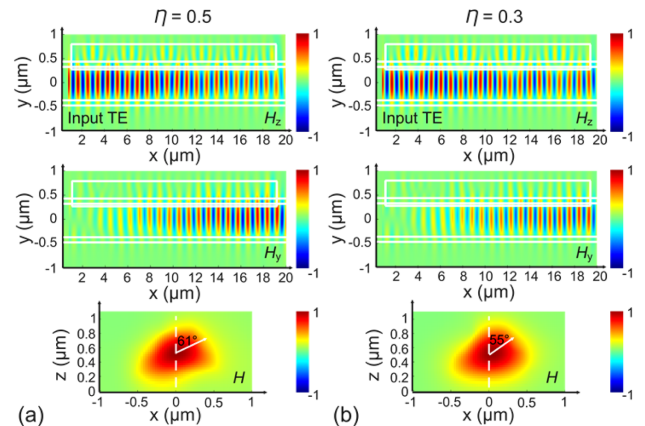


Fig. 11. Magnetic field evolution of the input TE mode through the proposed PR as the crystallization ratio of Sb_2Se_3 material is set as (a) $\eta = 0.5$ and (b) $\eta = 0.3$. The H_z component and H_y component are plotted, as well as the magnetic field distributions of the output modes.

to Fig. 8. Therefore, using the present device structure, we can achieve a tunable PR on the LNOI platform, which was not accomplished in previously reported in PRs with only one rotation function.

Finally, with the features of high CE ($>99\%$), low IL (<0.5 dB), compact size (device length of $17.7\ \mu\text{m}$), and tunable function, we believe the proposed LNOI-based PR can be an efficient polarization management device and that the reported tunable device scheme offers a good way to push the development of PICs based on the LNOI platform [4–8].

4. CONCLUSION

In conclusion, we proposed an efficient and tunable PR on the LNOI platform, in which the tunable polarization rotation is achieved by using low-loss phase change material Sb_2Se_3 . By depositing the Sb_2Se_3 layer on the designed LNOI waveguide in an asymmetrical way, the required asymmetrical structure for the polarization rotation is formed. Further, through optimizing the hybrid waveguide structure and setting the phase state of the Sb_2Se_3 layer to a crystalline state, we have achieved the efficient on-chip polarization rotation, where the obtained CE and IL are 99.6% (99.2%) and 0.38 dB (0.4 dB) for the TE-to-TM (TM-to-TE) rotation in a length of $17.7\ \mu\text{m}$. We should note that the polarization rotation angle can be varied for the present device only by changing the working state of the Sb_2Se_3 layer without affecting the device structure. The presence of such a tunable function is almost nil in previous reports. With these features, the proposed efficient PR can provide a tunable polarization manipulation way for the LNOI platform.

Funding. National Natural Science Foundation of China (62205129); Natural Science Foundation of Jiangsu Province (BK20200592); Fundamental Research Funds for the Central Universities (JUSRP12024).

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

REFERENCES

1. L. Arizmendi, "Photonic applications of lithium niobate crystals," *Phys. Status Solidi* **201**, 253–283 (2004).
2. E. L. Wooten, K. M. Kissa, A. Yi-Yan, E. J. Murphy, D. A. Lafaw, P. F. Hallemeier, D. Maack, D. V. Attanasio, D. J. Fritz, G. J. McBrien, and D. E. Bossi, "A review of lithium niobate modulators for fiber-optic communications systems," *IEEE J. Sel. Top. Quantum Electron.* **6**, 69–82 (2000).
3. F. Chen, "Photonic guiding structures in lithium niobate crystals produced by energetic ion beams," *J. Appl. Phys.* **106**, 081101 (2009).
4. D. Zhu, L. Shao, M. Yu, R. Cheng, B. Desiatov, C. J. Xin, Y. Hu, J. Holzgrafe, S. Ghosh, A. Shams-Ansari, E. Puma, N. Sinclair, C. Reimer, M. Zhang, and M. Lončar, "Integrated photonics on thin-film lithium niobate," *Adv. Opt. Photonics* **13**, 242–352 (2021).
5. J. Lin, F. Bo, Y. Cheng, and J. Xu, "Advances in on-chip photonic devices based on lithium niobate on insulator," *Photonics Res.* **8**, 1910–1936 (2020).
6. Y. Jia, L. Wang, and F. Chen, "Ion-cut lithium niobate on insulator technology: recent advances and perspectives," *Appl. Phys. Rev.* **8**, 011307 (2021).
7. Y. Qi and Y. Li, "Integrated lithium niobate photonics," *Nanophotonics* **9**, 1287–1320 (2020).
8. G. Chen, N. Li, J. D. Ng, H. Lin, Y. Zhou, Y. H. Fu, L. Y. T. Lee, Y. Yu, A. Liu, and A. J. Danner, "Advances in lithium niobate photonics: development status and perspectives," *Adv. Photonics* **4**, 034003 (2022).
9. T. Barwicz, M. R. Watts, M. A. Popović, P. T. Rakich, L. Socci, F. X. Kärtner, E. P. Ippen, and H. I. Smith, "Polarization-transparent microphotonic devices in the strong confinement limit," *Nat. Photonics* **1**, 57–60 (2007).
10. Z. Lin, Y. Lin, H. Li, M. Xu, M. He, W. Ke, H. Tan, Y. Han, Z. Li, D. Wang, X. S. Yao, S. Fu, S. Yu, and X. Cai, "High-performance polarization management devices based on thin-film lithium niobate," *Light Sci. Appl.* **11**, 93 (2022).
11. Z. Hou, X. Xiong, J. Cao, Q. Chen, Z. Tian, X. Ren, and H. Sun, "On-chip polarization rotators," *Adv. Opt. Mater.* **7**, 1900129 (2019).
12. L. Gao, Y. Huo, J. S. Harris, and Z. Zhou, "Ultra-compact and low-loss polarization rotator based on asymmetric hybrid plasmonic waveguide," *IEEE Photonics Technol. Lett.* **25**, 2081–2084 (2013).
13. H. Zhang, S. Das, J. Zhang, Y. Huang, C. Li, S. Chen, H. Zhou, M. Yu, P. G.-Q. Lo, and J. T. L. Thong, "Efficient and broadband polarization rotator using horizontal slot waveguide for silicon photonics," *Appl. Phys. Lett.* **101**, 021105 (2012).
14. J. Zhang, M. Yu, G.-Q. Lo, and D.-L. Kwong, "Silicon-waveguide-based mode evolution polarization rotator," *IEEE Sel. Top. Quantum Electron.* **16**, 53–60 (2010).
15. Y. Xu, J. Xiao, and X. Sun, "A compact hybrid plasmonic polarization rotator for silicon-based slot waveguides," *IEEE Photonics Technol. Lett.* **26**, 1609–1612 (2014).
16. W. Ye, X. Yuan, Y. Gao, and J. Liu, "Design of broadband silicon-waveguide mode-order converter and polarization rotator with small footprints," *Opt. Express* **25**, 33176–33183 (2017).
17. W. Chang, S. Xu, M. Cheng, D. Liu, and M. Zhang, "Inverse design of a single-step-etched ultracompact silicon polarization rotator," *Opt. Express* **28**, 28343–28351 (2020).
18. Z. Chen, J. Yang, W.-H. Wong, E. Y.-B. Pun, and C. Wang, "Broadband adiabatic polarization rotator-splitter based on a lithium niobate on insulator platform," *Photonics Res.* **9**, 2319–2324 (2021).
19. Y. Zhao, W. Yu, S. Dai, J. Li, T. Lin, M. Li, and J. Liu, "Compact lithium-niobate-on-insulator polarization rotator based on asymmetric hybrid plasmonics waveguide," *IEEE Photonics J.* **13**, 4800105 (2021).
20. J. Tao, X. Li, J. Li, S. Dai, Y. Zhao, C. Wei, and J. Liu, "Compact, high extinction ratio, and low-loss polarization beam splitter on lithium-niobate-on-insulator using a silicon nitride nanowire assisted waveguide and a grooved waveguide," *Photonics* **9**, 779 (2022).
21. C. Deng, M. Lu, Y. Sun, L. Huang, D. Wang, G. Hu, R. Zhang, B. Yun, and Y. Cui, "Broadband and compact polarization beam splitter in LNOI hetero-anisotropic metamaterials," *Opt. Express* **29**, 11627–11634 (2021).
22. H. Xu, D. Dai, L. Liu, and Y. Shi, "Proposal for an ultra-broadband polarization beam splitter using an anisotropy-engineered Mach-Zehnder interferometer on the x-cut lithium-niobate-on-insulator," *Opt. Express* **28**, 10899–10908 (2020).
23. M. Wuttig, H. Bhaskaran, and T. Taubner, "Phase-change materials for non-volatile photonic applications," *Nat. Photonics* **11**, 465–476 (2017).
24. Z. Fang, R. Chen, J. Zheng, and A. Majumdar, "Non-volatile reconfigurable silicon photonics based on phase-change material," *IEEE J. Sel. Top. Quantum Electron.* **28**, 8200317 (2022).
25. J. Wang, L. Wang, and J. Liu, "Overview of phase-change materials based photonic devices," *IEEE Access* **8**, 121211 (2020).
26. C. Wu, H. Yu, H. Li, X. Zhang, I. Takeuchi, and M. Li, "Low-loss integrated photonic switch using subwavelength patterned phase change material," *ACS Photonics* **6**, 87–92 (2019).
27. M. S. Nisar, X. Yang, L. Lu, J. Chen, and L. Zhou, "On-chip integrated photonic devices based on phase change materials," *Photonics* **8**, 205 (2021).
28. F. Bruckerhoff-Pluckelmann, J. Feldmann, C. D. Wright, H. Bhaskaran, and W. H. P. Pernice, "Chalcogenide phase-change devices for neuromorphic photonic computing," *J. Appl. Phys.* **129**, 151103 (2021).
29. J. Feldmann, N. Youngblood, C. D. Wright, H. Bhaskaran, and W. H. P. Pernice, "All-optical spiking neurosynaptic networks with self-learning capabilities," *Nature* **569**, 208–214 (2019).

30. C. Lian, C. Vagionas, T. Alexoudi, N. Pleros, N. Youngblood, and C. Ríos, "Photonic (computational) memories: tunable nanophotonics for data storage and computing," *Nanophotonics* **11**, 3823–3854 (2022).
31. M. Delaney, I. Zeimpekis, D. Lawson, D. W. Hewak, and O. L. Muskens, "A new family of ultralow loss reversible phase-change materials for photonic integrated circuits: Sb_2S_3 and Sb_2Se_3 ," *Adv. Fun. Mater.* **30**, 2002447 (2020).
32. M. Delaney, I. Zeimpekis, H. Du, X. Yan, M. Banakar, D. J. Thomson, D. W. Hewak, and O. L. Muskens, "Nonvolatile programmable silicon photonics using an ultralow-loss Sb_2Se_3 phase change material," *Sci. Adv.* **7**, eabg3500 (2021).
33. C. L. Xu, W. P. Huang, M. S. Stern, and S. K. Chaudhuri, "Full-vectorial mode calculations by finite difference method," *IEEE Proc. Optoelectron.* **141**, 281–286 (1994).
34. Lumerical Mode Solutions, <https://www.ansys.com/products/photonics/mode>.
35. Lumerical FDTD Solutions, <https://www.ansys.com/products/photonics/fdtd>.
36. E. D. Palik, *Handbook of Optical Constants of Solids* (Academic, 1998).
37. R. R. Ghosh and A. Dhawan, "Numerical modeling of an integrated non-volatile reflector switch and mode converter switch based on a low loss phase change material (Sb_2Se_3) in SiN platforms," *Opt. Mater. Express* **12**, 4268–4283 (2022).